THE LEVEL FIELD
Admissions Intelligence Series

QUEEN ELIZABETH'S SCHOOL

Distractors In Maths

GL Assessment Multiple-Choice Architecture and Engineered Traps

3	4–6	235+
Primary Distractor Snares	Preventable Errors Per Paper	SAS Target Score

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What This Document Is For

Standard 11+ practice materials condition candidates to believe that multiple-choice options consist of one correct answer and four arbitrary numbers. This is a fatal assumption.

In the QE Boys test, answer options are mathematically engineered psychological traps. The bespoke GL Assessment psychometricians calculate the exact numerical outputs of a 10-year-old's most common procedural errors, missed unit conversions, and half-finished workings. This document deconstructs each snare type and provides the mechanical protocol to neutralise them.

> *"To secure a 235+ Standardised Age Score, candidates must anticipate the three primary distractor snares."*

01 The QE Boys Multiple-Choice Architecture

Engineered Traps

In the QE Boys test, answer options are mathematically engineered psychological traps. The psychometricians do not use random numbers: they calculate the exact numerical outputs of a 10-year-old's most common procedural errors, missed unit conversions, and half-finished workings.

5

OPTIONS PER QUESTION

One correct answer. Four engineered distractors, each corresponding to a specific procedural error. Under the 50-minute time constraint, these traps induce a false sense of security, causing highly capable candidates to confidently select incorrect answers.

HOW DISTRACTORS ARE BUILT

GL Assessment psychometricians reverse-engineer the most statistically probable errors for each question type. Every distractor corresponds to a real computational path a stressed 10-year-old will follow.

WHY CAPABLE CANDIDATES FAIL

Under severe time pressure, a familiar-looking number in the options list triggers premature relief. The candidate stops calculating, bubbles the OMR sheet, and moves on. The error is invisible until marking.

02 Distractor Type I: The Intermediate Calculation Snare

Premature Relief

The QE Boys mathematics paper relies heavily on complex word problems requiring three or four distinct computational steps. Exam writers intentionally calculate the exact numerical value of Step 1 or Step 2 and insert it directly into Option A or Option B.

A. THE MECHANISM

An anxious candidate executes the first difficult step, spots their resulting number waiting in the options, experiences a dopamine rush of premature relief, bubbles the OMR sheet, and abandons the final calculation steps entirely.

QE Boys Application: A problem asks for the final sale price of a £400 item after a 15% discount and an added £10 delivery fee. Option A will be £60 (trap: the discount amount). Option B will be £340 (trap: forgetting the delivery fee). Option C will be £350 (the correct answer).

The Option Distribution

For the £400 discount problem above, the answer options are structured as follows. Each incorrect option corresponds to a specific incomplete execution path.

Option	What It Traps	Status
A: £60	Candidate stops at Step 1 (the discount amount)	Distractor
B: £340	Candidate completes Steps 1 and 2, forgets delivery fee	Distractor
C: £350	All three steps completed correctly	**Correct Answer**

03 Distractor Type II: The Unit Inversion Snare

Unit Blindness

Answer options frequently contain the mathematically correct integer, but paired with an incorrect, unconverted unit of measurement. GL Assessment weaponises this by presenting mixed units within the stimulus paragraph.

THE SNARE

A candidate correctly calculates an area as 120, but fails to convert from centimetres to metres. The options will include 120m², 1.2cm², and 1.2m². The correct integer paired with the wrong unit catches every rushing candidate.

TACTICAL DEFENCE

The Circle-and-Sync Rule: physically circle all units of measurement within the prompt before initiating calculations. If the prompt uses mixed units, all data points must be converted to the required output unit immediately.

04 Distractor Type III: The Subtractive Reversal Snare

The Inverted Output

GL Assessment structures questions to deliver a dense block of data, leading the candidate down an assumed logical path, only to invert the required output at the very end of the paragraph.

B. THE PSYCHOLOGICAL SNARE

The candidate calculates the most obvious data point. However, the QE test asks for the inverse. The incorrect assumption will be waiting as a distractor option.

QE Boys Application: A paragraph outlines the ratios of marbles between three boys. The assumed question is how many marbles does Boy A have. The actual question asks how many more marbles does Boy A have than Boy C. The individual totals for all three boys will all be explicitly listed as distractor options.

05 Tactical Execution: The Terminal Sentence Protocol

The Mechanical Override

In the QE Boys test, the actual directive is always buried or inverted in the final sentence. Before the pencil touches the OMR bubble sheet, the candidate must physically isolate and heavily underline the terminal clause of the mathematical paragraph.

The Two Strategic Protocols

To systematically neutralise all three distractor traps, candidates must deploy two interlocking mechanical protocols for every mathematical word problem in the examination hall.

1

The Terminal Sentence Protocol

- Before beginning any calculation, physically locate the final sentence of the word problem.
- Underline it heavily. Identify the exact output required: a discount, a final price, a sum, a difference, centimetres, or metres.
- Only then begin calculation. This single 3-second check forces executive function to verify the correct output before any arithmetic begins.

Note: This protocol addresses all three distractor types simultaneously. It prevents the Intermediate Calculation Snare, forces unit verification, and catches the Subtractive Reversal in a single mechanical step.

2

The Circle-and-Sync Rule

- Physically circle every unit of measurement mentioned in the problem before touching any numbers.
- If mixed units appear in the same problem, convert all data points to the required output unit before executing any calculation.
- Sync the circled output unit from the terminal sentence against the unit in the answer options before bubbling. A numeric match without a unit match is not a correct answer.

Note: These two protocols together prevent 4 to 6 highly preventable distractor errors per paper. That alone represents the scoring margin between a waitlist result and an unconditional offer.